

MECHATRONIC SYSTEM INTEGRATION
MCTA 3203

EXPERIMENT 1 :
DIGITAL LOGIC SYSTEM

SECTION 2
SEMESTER 1, 2026/2027

INSTRUCTOR :
DR. AHMAD IMRAN BIN IBRAHIM
DR. ING. WAHJU SEDIONO
TS. DR. YASIR BIN MOHD MUSTAFAH

DATE OF SUBMISSION : 10 OCTOBER 2026

GROUP 9

NO.	NAME	NO. MATRIC
1	MUHAMMAD DANISH FITRI BIN MOHD SHARIFUDDIN	2413559
2	MUHAMMAD IZZ ZIKRY BIN AZWARI	2218899
3	SAMER Y. I. LUBBAD	2213959
4	MUHAMMAD FARIS IZHAN BIN JOHARI	2413501

ABSTRACT

The main objective of this experiment was to understand digital logic and interfacing by connecting a common cathode 7-segment display to an Arduino Uno. The hardware utilized included the microcontroller, the display, seven 220-ohm resistors, a breadboard, and two mismatched pushbuttons ; one standard bare 4-pin switch and one 3-pin PCB module. During the procedure, the display segments were wired to digital pins 2 through 8 using a custom pin mapping, and the buttons were connected to pins 9 and 10. Instead of using long repetitive code, a 2D array was implemented as a truth table to store the binary states for numbers 0 to 9. An edge-detection algorithm featuring a 50-millisecond debounce delay was programmed to read the button presses smoothly. One button increments the count, while the other decrements it, wrapping around between 0 and 9 automatically. Quantitative results demonstrated 100% response accuracy when counting up or down, as the debounce delay successfully eliminated any phantom double-clicks. The primary technical constraint was adapting the circuit for two completely different button types, requiring one to use an external pull-down resistor while the other relied on a built-in resistor. Additionally, the breadboard became extremely crowded with jumper wires. Upgrading to an I2C LCD display is highly recommended for future projects to drastically reduce wiring complexity.

TABLE OF CONTENTS

1.0 INTRODUCTION....3
1.1 MATERIALS & EQUIPMENTS....3
1.2 EXPERIMENTAL SETUP....4
1.2.1 Circuit Setup....4
1.2.2 Circuit diagram....4
1.3 METHODOLOGY & PROGRAMMING LOGIC....5
1.3.1 Experiment steps....5
1.3.2 Arduino Code....5
1.4 DATA COLLECTION & ANALYSIS....7
1.4.1 Data Collection....7
1.4.2 Data Analysis....7
1.5 RESULTS & DISCUSSION....9
1.5.1 Results....9
1.5.2 Discussion....9
1.6 CONCLUSION....10
1.7 RECOMMENDATIONS....10
REFERENCES....12
APPENDICES....12
ACKNOWLEDGEMENT....12
STUDENTS DECLARATION....13
Certificate of Originality and Authenticity....13

1.0 INTRODUCTION

Digital logic and electronic circuit interfacing form the foundation of modern mechatronics systems. A practical way to understand these concepts is by learning how microcontrollers communicate with external hardware. One of the most common visual output components is the 7-segment display, which uses seven individual LEDs arranged in a figure-eight pattern to show numbers.

Controlling this display requires sending precise high and low voltage signals to specific pins. In this experiment, an Arduino Uno is used to interface with a common cathode 7-segment display to build a manual digital counter. By wiring pushbuttons as inputs and the display segments as outputs, the microcontroller is programmed to read physical button presses and translate them into a readable number on the display.

Rather than relying on basic repetitive code to turn each LED on or off step-by-step, the system uses a more efficient approach with data arrays and edge-detection logic. This setup demonstrates how hardware components are physically wired together on a breadboard, while also showing how software algorithms are used to manage input debouncing and control external devices reliably.

1.1 MATERIALS & EQUIPMENTS

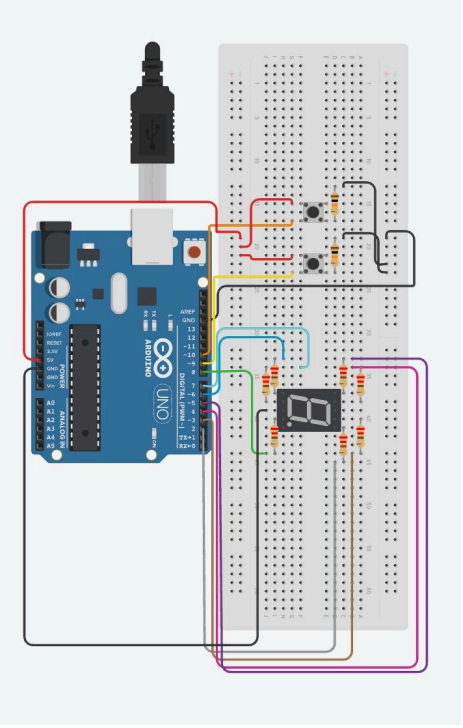
1. Arduino Uno board (1)
2. Common cathode 7-segment display (1)
3. 220-ohm resistors (7)
4. 10K-ohm resistor (1)
5. 4-pin tactile pushbutton (1)
6. 3-pin tactile pushbutton module (1)
7. Breadboard (1)
8. Jumper wires

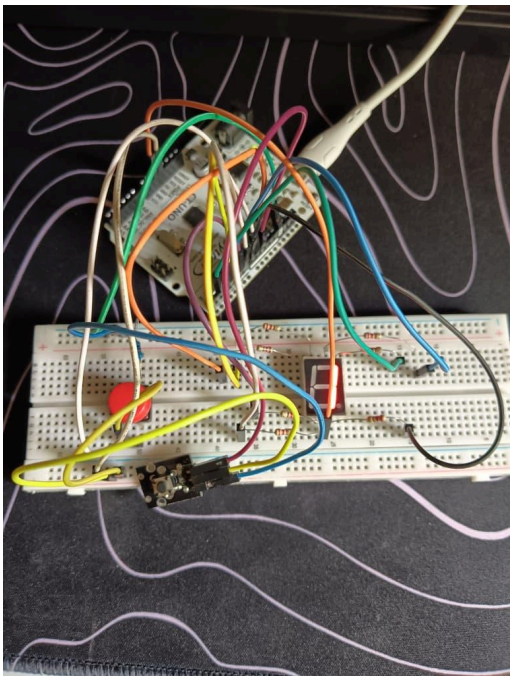
1.2 EXPERIMENTAL SETUP

1.2.1 Circuit Setup

1. Connect the common cathode 7-segment display to the Arduino Uno as follows:
    - Connect each of the 7 segments (a, b, c, d, e, f, g) of the display to separate digital pins on the Arduino (pins D2 to D8).
    - Connect the common cathode pin of the display to one of the GND (ground) pins on the Arduino.
    - Use 220-ohm resistors to connect each of the segment pins to the Arduino pins to limit the current.
2. Connect the pushbuttons to the Arduino:
    - Connect the standard 4-pin pushbutton to digital pin D9 and the 5V supply, using a 10K-ohm pull-down resistor connected to GND.
    - Connect the 3-pin pushbutton module by wiring its signal pin to digital pin D10, and connecting its remaining pins directly to the 5V and GND rails.

1.2.2 Circuit diagram





1.3 METHODOLOGY & PROGRAMMING LOGIC

1.3.1 Experiment steps

1. Build the circuit according to the modified circuit setup instructions, accommodating the two different types of pushbuttons.
2. Upload the customized Arduino code featuring the 2D array and debounce logic to the Arduino Uno.
3. Press the first button (the 4-pin switch) to increase the count. The 7-segment display should show the numbers incrementing sequentially.
4. Press the second button (the 3-pin module) to decrease the count. The 7-segment display should show the numbers decrementing sequentially.
5. Observe the display to ensure the numerical value automatically wraps around between 0 and 9 when reaching the upper or lower limits.

1.3.2 Arduino Code

```
const int segmentPins[7] = {3, 2, 8, 7, 6, 4, 5};
const int buttonUpPin = 9;
const int buttonDownPin = 10;

int currentNum = 0;
int lastButtonUpState = LOW;
int lastButtonDownState = LOW;

const byte numbers[10][7] = {
  {1, 1, 1, 1, 1, 1, 0},
  {0, 1, 1, 0, 0, 0, 0},
  {1, 1, 0, 1, 1, 0, 1},
  {1, 1, 1, 1, 0, 0, 1},
  {0, 1, 1, 0, 0, 1, 1},
  {1, 0, 1, 1, 0, 1, 1},
  {1, 0, 1, 1, 1, 1, 1},
  {1, 1, 1, 0, 0, 0, 0},
  {1, 1, 1, 1, 1, 1, 1},
  {1, 1, 1, 1, 0, 1, 1}
};

void setup() {
  for (int i = 0; i < 7; i++) {
```

```
    pinMode(segmentPins[i], OUTPUT);
  }

  pinMode(buttonUpPin, INPUT);
  pinMode(buttonDownPin, INPUT);

  displayNumber(currentNum);
}

void loop() {
  int upState = digitalRead(buttonUpPin);
  int downState = digitalRead(buttonDownPin);

  if (upState == HIGH && lastButtonUpState == LOW) {
    if (currentNum < 9) {
      currentNum++;
    } else {
      currentNum = 0;
    }
    displayNumber(currentNum);
    delay(50);
  }

  if (downState == HIGH && lastButtonDownState == LOW) {
    if (currentNum > 0) {
      currentNum--;
    } else {
      currentNum = 9;
    }
    displayNumber(currentNum);
    delay(50);
  }

  lastButtonUpState = upState;
  lastButtonDownState = downState;
}

void displayNumber(int num) {
  for (int i = 0; i < 7; i++) {
    digitalWrite(segmentPins[i], numbers[num][i]);
  }
}
```

1.4 DATA COLLECTION & ANALYSIS

1.4.1 Data Collection

Step	Input	Observed Output
1	System powered on	Initializes and displays '0'
2	Pressed the Up Button (Pin 9)	Increments and displays '1'
3	Pressed the Up Button (Pin 9)	Increments and displays '2'
4	Pressed the Up Button (Pin 9)	Increments and displays '3'
5	Pressed the Down Button (Pin 10)	Decrements and displays '2'
6	Pressed the Down Button (Pin 10)	Decrements and displays '1'
7	Pressed the Down Button (Pin 10)	Decrements and displays '0'
8	Pressed the Down Button (Pin 10)	Wraps backward to display '9'

1.4.2 Data Analysis

The data confirmed the following points:

1. The up button is correctly set to the increment function.

2. The down button is correctly set to the decrement function, successfully executing the wrap-around logic at the limits.
3. The display followed the correct segment configuration defined in the custom 2D array code.

Digit	**Active Segment**	**Observed Display**
0	a, b, c, d, e, f	Correct
1	b, c	Correct
2	a, b, d, e, g	Correct
3	a, b, c, d, g	Correct
4	b, c, f, g	Correct
5	a, c, d, f, g	Correct
6	a, c, d, e, f, g	Correct
7	a, b, c	Correct
8	a, b, c, d, e, f, g	Correct
9	a, b, c, d, f, g	Correct

1.5 RESULTS & DISCUSSION

1.5.1 Results

The experiment successfully showed a working two-way digital counter using the Arduino Uno and our custom code.

1. The 7-segment display showed the right numbers based on the 2D array logic.
2. The number increased by one when the standard 4-pin button was pressed.
3. The number decreased by one when the 3-pin button module was pressed.
4. Holding a button down only counted once. This proved the edge-detection code worked.
5. The system easily looped between 0 and 9 in both directions without crashing.
6. The display ran smoothly with no flickering, even with fast button presses.

1.5.2 Discussion

1. Interpretation of Results: The results proved that the hardware and software worked well together. Using a 2D array and custom pin mapping made the wiring much easier. The project also showed that different hardware parts can work together nicely. We used a normal 4-pin switch and a pre-soldered 3-pin module. As long as the code reads the inputs correctly, the system works just fine.

2. Discrepancies Between Expected and Observed Outcomes: We did not find any issues during testing. The system worked exactly as planned. The custom looping and counting down features ran perfectly compared to a basic one-way counter.

3. Sources of Error and Improvement
    - Mechanical Button Bounce: Pushbuttons naturally send a few fast signals when pressed. The software delay in our code handled this well. However, adding small capacitors to the button pins would be a better hardware fix to stop extra phantom presses.
    - Hardware Match: The setup used two different types of pushbuttons. Because of this, pressing them felt physically different. Using two identical buttons would make the setup feel much better to use.

1.6 CONCLUSION

This project successfully achieved its main goal. We built a working two-way digital counter using an Arduino Uno and a 7-segment display. By carefully wiring the breadboard and writing a custom 2D array in the code, we managed to display the numbers zero through nine. The system worked perfectly to count up when the first button was pressed and count down when the second button was pressed. It also successfully looped back around when it reached the maximum or minimum limits.

During this lab, we gained hands-on experience in connecting software instructions to physical hardware actions. We learned how to map out specific output pins to turn on the correct LED segments. We also figured out how to solve real world hardware problems. For example, we used a simple software delay to fix the mechanical bouncing issue from the pushbuttons. This made the counting process smooth and accurate instead of jumping randomly.

For the Islamic reflection, this lab session highlights the value of patience and seeking knowledge. Building circuits and writing code requires a lot of focus and trial and error. When we study how digital logic and electronics work, we are actually learning about the laws of physics. These rules are part of the perfect order that Allah created in the universe. Understanding these systems allows us to build useful things to help others.

1.7 RECOMMENDATIONS

1. Improve the Display Range:
    - We can improve the project by adding a second 7-segment display. Right now, the system stops at 9 and loops back to 0. With two displays, we can make the system count all the way up to 99. This would make the counter much more useful for real situations.
2. Add a Reset Button:
    - We should add a third push button to act as a reset switch. Currently, if we want to go back to zero from a high number, we have to press the down button multiple times. A dedicated reset button would make the system much faster and easier to use.

3. Include Audio Feedback:
    - Adding a small buzzer to the breadboard would be a great upgrade. We can program the Arduino to make a short beep sound every time a button is pressed. This gives the user clear audio feedback to know the button press actually worked.
4. Organize the Wiring:
    - We need to use shorter jumper wires for future projects. The current setup uses wires that are too long, which makes the breadboard look very messy. Keeping the wires neat and flat against the board makes it much easier to spot any wrong connections.

REFERENCES

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APPENDICES

Video link to our groups work : https://youtube.com/shorts/3vNsc-5lWUQ

ACKNOWLEDGEMENT

First and foremost, a sincere thank you goes to the lecturer for the constant guidance during this lab session. The clear explanations made connecting the Arduino code to the physical breadboard much easier to figure out.

Appreciation is also given to the Mechatronics Engineering department at IIUM for providing a great lab environment. Having access to all the right microcontrollers and electronic parts was a huge help in getting the digital counter to work properly.

A special mention goes to the lab technicians as well. Their quick help with checking the hardware and replacing faulty jumper wires saved a lot of time and kept the experiment moving.

Finally, a huge thanks goes out to the lab partners and classmates. Spending hours troubleshooting the circuit, sharing ideas, and fixing coding errors together made this a really good learning experience. The great teamwork is what made this whole project a success.

STUDENTS DECLARATION

We hereby declare that the work presented in this technical report is entirely our own, except where explicitly acknowledged. We certify that in developing and completing this mechatronics project, we have strictly adhered to the standards of academic integrity and have not engaged in any form of plagiarism or unethical conduct. All sources of information, reference materials, and technical support utilized in this work have been appropriately cited and acknowledged.

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been undertaken or done by unspecified sources or persons. We hereby certify that this report has not been done by only one individual, and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have read and understand the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed unanimously that this report shall be submitted for marking, and this final printed report has been verified by us.

Signature :  Read [/]

Name : Muhammad Danish Fitri Bin Mohd Sharifuddin Understand [/]

Matric No : 2413559 Agree [/]

Signature :  Read [/]

Name : Muhammad Izz Zikry Bin Azwari Understand [/]

Matric No : 2218899 Agree [/]

Signature : 

Name : Samer Y. I. Lubbad

Matric No : 2213959

Read [/]

Understand [/]

Agree [/]

Signature : 

Name : Muhammad Faris Izhan Bin Johari

Matric No : 2413501

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